Chapter 1. Groups and Subgroups

Sets and relations, binary operations, isomorphisms, groups, subgroups, cyclic groups, generating sets

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These notes follow Fraleigh, *A First Course in Abstract Algebra*; the book's numbering is given in brackets.

1.1 Sets and relations (§0)

Motivation. Cosets, factor groups and residue classes are sets of equivalence classes; Theorem 0.22 (equivalence relation = partition) makes such constructions legitimate.

A **relation** between sets A and B is a subset $\mathcal{R} \subset A \times B$; a **function** $\phi : A \to B$ is a relation in which each $a \in A$ appears as first member of exactly one pair (a, b), written $b = \phi(a)$. ϕ is **one-to-one** (injective) if $\phi(a_1) = \phi(a_2)$ implies $a_1 = a_2$, **onto** (surjective) if every b is a value, and a **bijection** if both. Sets have the same **cardinality** if there is a bijection between them. A **partition** of S is a collection of nonempty disjoint subsets (**cells**) whose union is S. A relation $\sim$ on S is an **equivalence relation** if it is reflexive ($x \sim x$), symmetric ($x \sim y \Rightarrow y \sim x$) and transitive ($x \sim y$, $y \sim z \Rightarrow x \sim z$).

Theorem 1.1 (0.22, equivalence relations and partitions)

Let $\sim$ be an equivalence relation on a nonempty set S. The sets $\bar{a} = \{x \in S : x \sim a\}$ form a partition of S. Conversely, each partition of S gives the equivalence relation "$x \sim y$ iff x, y lie in the same cell".

Proof. Each a lies in $\bar{a}$ (reflexivity), so the cells are nonempty and cover S. Suppose $c \in \bar{a} \cap \bar{b}$. If $x \in \bar{a}$, then $x \sim a$, $a \sim c$ (symmetry) and $c \sim b$, so $x \sim b$ (transitivity) and $\bar{a} \subset \bar{b}$; symmetrically $\bar{b} \subset \bar{a}$. So distinct cells are disjoint.

Conversely, every x is in a cell with itself; if x, y share a cell so do y, x; and if x, y share a cell and y, z share a cell, these cells contain y and hence coincide. ■

Example 1.1

Congruence modulo n: for $n \in \mathbb{Z}^+$, $h \equiv k \pmod{n}$ iff n divides $h - k$. It is an equivalence relation whose cells are the n **residue classes modulo** n.

1.2 Binary operations and isomorphisms (§§1–3)

Motivation. Isomorphism makes "behaves identically" precise, as for $\langle \mathbb{R}, + \rangle \cong \langle \mathbb{R}^+, \cdot \rangle$ via $x \mapsto e^x$; structures are told apart by a structural property one has and the other lacks.

A **binary operation** $*$ on S is a function $S \times S \to S$. It is **commutative** if $a * b = b * a$ and **associative** if $(a * b) * c = a * (b * c)$ for all a, b, c. A subset H is **closed** under $*$ if $a * b \in H$ for $a, b \in H$. Examples: $+$ and $\cdot$ on $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$; on the circle $U = \{z \in \mathbb{C} : |z| = 1\}$, multiplication; on $\mathbb{R}_c = [0, c)$, addition modulo c; on $U_n = \{z : z^n = 1\}$ and $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$, multiplication and addition modulo n.

Theorem 1.2 (2.13, associativity of composition)

For functions $f, g, h : S \to S$, $f \circ (g \circ h) = (f \circ g) \circ h$.

Proof. For $x \in S$, both sides send x to $f(g(h(x)))$. ■

A **binary algebraic structure** $\langle S, * \rangle$ is a set with a binary operation. An **isomorphism** $\phi : \langle S, * \rangle \to \langle S', *' \rangle$ is a bijection with $\phi(x * y) = \phi(x) *' \phi(y)$. To show two structures are isomorphic: define ϕ, show it is one-to-one and onto, and verify the homomorphism property. To show they are not, find a **structural property** (preserved by every isomorphism, such as cardinality, commutativity, or existence of solutions of $x * x = c$) held by one and not the other.

Example 1.2

$\phi(x) = e^x$ is an isomorphism $\langle \mathbb{R}, + \rangle \to \langle \mathbb{R}^+, \cdot \rangle$. $\langle \mathbb{R}_{2\pi}, +_{2\pi} \rangle \cong \langle U, \cdot \rangle$ via $x \mapsto e^{ix}$, and $\langle \mathbb{Z}_n, +_n \rangle \cong \langle U_n, \cdot \rangle$ via $k \mapsto e^{2\pi i k/n}$. $\langle \mathbb{Q}, + \rangle \ncong \langle \mathbb{R}, + \rangle$ (cardinality), $\langle \mathbb{Z}, + \rangle \ncong \langle \mathbb{Z}^+, + \rangle$ (in $\mathbb{Z}$, $x + x = c$ has a solution for $c = 0$), and $\langle \mathbb{R}, \cdot \rangle \ncong \langle \mathbb{C}, \cdot \rangle$ ($x \cdot x = c$ is solvable for every $c \in \mathbb{C}$).

An **identity element** e satisfies $e * s = s * e = s$ for all s; it is unique ($e = e * e' = e'$).

Theorem 1.3 (3.14)

If $\langle S, * \rangle$ has an identity e and $\phi : \langle S, * \rangle \to \langle S', *' \rangle$ is an isomorphism, then $\phi(e)$ is an identity for $*'$.

Proof. For $s' \in S'$ write $s' = \phi(s)$ (onto). Then $\phi(e) *' s' = \phi(e * s) = \phi(s) = s'$, and similarly $s' *' \phi(e) = s'$. ∎

1.3 Groups (§4)

Motivation. A group abstracts an operation in which every equation $a * x = b$ is uniquely solvable: associativity, identity, inverses.

Definition (Group)

A **group** $\langle G, * \rangle$ is a set with a binary operation such that
$\mathcal{G}_1$: $*$ is associative;
$\mathcal{G}_2$: there is $e \in G$ with $e * x = x * e = x$ for all x;
$\mathcal{G}_3$: each a has an inverse a' with $a' * a = a * a' = e$.
It is **abelian** if $*$ is commutative.

Theorem 1.4 (4.15, cancellation laws)

In a group, $a * b = a * c$ implies $b = c$, and $b * a = c * a$ implies $b = c$.

Proof. Multiply $a * b = a * c$ on the left by a' and use associativity: $b = e * b = (a' * a) * b = a' * (a * b) = a' * (a * c) = c$. The right law is symmetric. ∎

Theorem 1.5 (4.16)

In a group, the equations $a * x = b$ and $y * a = b$ have unique solutions x, y.

Proof. $x = a' * b$ is a solution, and any two solutions satisfy $a * x_1 = a * x_2$, hence are equal by Theorem 1.4. Similarly $y = b * a'$. ∎

Theorem 1.6 (4.17)

In a group the identity element is unique, and each element has a unique inverse.

Proof. Identity: $e = e * e' = e'$. If a' and a'' are inverses of a, then $a * a' = e = a * a''$, so $a' = a''$ by cancellation. ■

We write a^{-1} for the inverse and, multiplicatively, a^n for powers ($a^0 = e$, $a^{-n} = (a^{-1})^n$); additively, $-a$ and na.

Corollary 1.7 (4.18)

$(a * b)' = b' * a'$.

Proof. $(a*b)*(b'*a') = a*(b*b')*a' = a*a' = e$, so $b'*a'$ is the inverse of $a*b$ by uniqueness. ■

Example 1.3

$\langle \mathbb{Z}, + \rangle$, $\langle \mathbb{Q}^*, \cdot \rangle$, $\langle \mathbb{Z}_n, +_n \rangle$, $\langle U_n, \cdot \rangle$, and $GL(n, \mathbb{R})$ (invertible $n \times n$ matrices under multiplication, nonabelian for $n \geq 2$) are groups; $\langle \mathbb{Z}^+, + \rangle$ and $\langle \mathbb{Z}, \cdot \rangle$ are not. Up to isomorphism there is exactly one group of each order $1, 2, 3$ (the group table is forced: each element appears exactly once in each row and column, by Theorem 1.5), and two groups of order 4: $\mathbb{Z}_4$ and the Klein 4-group V, in which every element is its own inverse.

1.4 Subgroups (§5)

Motivation. A group is studied through its subgroups; the most important source is the powers of a single element.

$H \subset G$ is a **subgroup**, $H \leq G$, if it is closed under the operation of G and is a group under it. The **order** $|G|$ of G is its number of elements.

Theorem 1.8 (5.14)

$H \subset G$ is a subgroup iff (1) H is closed under the operation, (2) $e \in H$, and (3) $a \in H \Rightarrow a^{-1} \in H$.

Proof. If $H \leq G$, (1) holds by definition. The identity e_H of H satisfies $e_H e_H = e_H = e_H e$ in G, so $e_H = e$ by cancellation, giving (2); the inverse of a in H is an inverse in G, hence equals a^{-1}, giving (3). Conversely (1)–(3) give $\mathcal{G}_2$ and $\mathcal{G}_3$ in H, and associativity is inherited from G. ■

For finite H, closure alone suffices (Exercise in the book: the powers of a repeat, so a^{-1} is a power of a). Equivalently, a nonempty H is a subgroup iff $ab^{-1} \in H$ for all $a, b \in H$.

Theorem 1.9 (5.17)

For $a \in G$, $H = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G, and it is the smallest subgroup containing a.

Proof. $a^r a^s = a^{r+s}$, $e = a^0$, $(a^r)^{-1} = a^{-r}$; apply Theorem 1.8. Any subgroup containing a contains all a^n by closure under products and inverses. ■

$H = \langle a \rangle$ is the **cyclic subgroup generated by** a; G is **cyclic** if $G = \langle a \rangle$ for some a, a **generator**. The **order** of a is $|\langle a \rangle|$.

1.5 Cyclic groups (§6)

Motivation. Cyclic groups are classified completely: each is $\mathbb{Z}$ or some $\mathbb{Z}_n$, and their subgroups are cyclic.

Theorem 1.10 (6.1)

Every cyclic group is abelian.

Proof. $a^r a^s = a^{r+s} = a^s a^r$. ■

Theorem 1.11 (6.3, division algorithm for $\mathbb{Z}$)

If $m \in \mathbb{Z}^+$ and $n \in \mathbb{Z}$, there are unique $q, r \in \mathbb{Z}$ with $n = mq + r$ and $0 \le r < m$.

Proof. The set $\{n - mq : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty (take $q = -|n|$), so has a least element $r = n - mq$. If $r \ge m$, then $r - m = n - m(q+1) \ge 0$ is smaller.

Uniqueness: if $mq + r = mq' + r'$ with $0 \le r, r' < m$, then m divides $r - r'$ and $|r - r'| < m$, so $r = r'$ and $q = q'$. ■

Theorem 1.12 (6.6)

A subgroup of a cyclic group is cyclic.

Proof. Let $G = \langle a \rangle$ and $H \le G$. If $H = \{e\}$, $H = \langle e \rangle$. Otherwise H contains some a^n with $n > 0$ (if $a^{-n} \in H$, so is a^n); let m be the least such. For $b = a^n \in H$ write $n = mq + r$, $0 \le r < m$; then $a^r = a^n (a^m)^{-q} \in H$, so $r = 0$ by minimality, and $b = (a^m)^q$. Thus $H = \langle a^m \rangle$. ■

Corollary 1.13 (6.7)

The subgroups of $\langle \mathbb{Z}, + \rangle$ are exactly the groups $n\mathbb{Z}$, $n \in \mathbb{Z}$.

Proof. $\mathbb{Z} = \langle 1 \rangle$; a subgroup generated by n is $n\mathbb{Z}$. ■

For $r, s \in \mathbb{Z}^+$ the subgroup $\{nr + ms : n, m \in \mathbb{Z}\}$ is $d\mathbb{Z}$ for a unique $d > 0$, the **greatest common divisor** $\gcd(r, s)$: d divides r and s, every common divisor divides $d = nr + ms$, and d is a linear combination of r and s. r, s are **relatively prime** if $\gcd(r, s) = 1$. If r, s are relatively prime and r divides sm, then r divides m (multiply $1 = ar + bs$ by m).

Theorem 1.14 (6.10)

Let $G = \langle a \rangle$. If G is infinite, $G \cong \langle \mathbb{Z}, + \rangle$; if $|G| = n$, $G \cong \langle \mathbb{Z}_n, +_n \rangle$.

Proof. If all powers of a are distinct ($a^h = a^k$, $h > k$ would give $a^{h-k} = e$), then $a^m \mapsto m$ is a well-defined bijection $G \to \mathbb{Z}$, and $a^i a^j = a^{i+j}$ shows it is an isomorphism. Otherwise let n be the least positive integer with $a^n = e$. Then $e, a, \dots, a^{n-1}$ are distinct (else a smaller positive power is e), and for any $m = nq + r$, $a^m = a^r$ (Theorem 1.11); so $G = \{a^0, \dots, a^{n-1}\}$, $|G| = n$, and $a^i \mapsto i$ is an isomorphism with $\mathbb{Z}_n$, since $a^i a^j = a^{i +_n j}$. ■

In particular, if a has finite order n, then $a^m = e$ iff n divides m.

Theorem 1.15 (6.14)

Let $G = \langle a \rangle$ have order n and $b = a^s$. Then b generates a cyclic subgroup of order n/d, $d = \gcd(n, s)$. Also $\langle a^s \rangle = \langle a^t \rangle$ iff $\gcd(s, n) = \gcd(t, n)$.

Proof. $|\langle b \rangle|$ is the least $m > 0$ with $a^{sm} = e$, i.e. with $n \mid sm$. Write $n = dn'$, $s = ds'$ with $\gcd(n', s') = 1$; then $n \mid sm$ iff $n' \mid s'm$ iff $n' \mid m$. So the order is $n' = n/d$. For the last statement: in the cyclic group $\mathbb{Z}_n$, $\langle s \rangle$ is the unique subgroup of order n/d (it is $\langle d \rangle$: $d \in \langle s \rangle$ as $d = xs + yn$, and $\langle d \rangle$ has order n/d); so $\langle a^s \rangle = \langle a^{\gcd(s,n)} \rangle$, and distinct divisors d of n give subgroups of distinct orders. ■

Corollary 1.16 (6.16)

If a generates a cyclic group of order n, the other generators are the a^r with r relatively prime to n.

Proof. a^r generates iff its order $n/\gcd(r, n)$ is n. ■

Consequently a cyclic group of order n has exactly one subgroup of each order d dividing n, and $\phi(n)$ generators.

1.6 Generating sets and Cayley digraphs (§7)

Motivation. Most groups we meet are generated by a few elements; the Cayley digraph pictures a group element as a path.

Theorem 1.17 (7.4)

The intersection of any family of subgroups H_i $(i \in I)$ of G is a subgroup of G.

Proof. If $a, b \in \bigcap H_i$, then ab and a^{-1} lie in every H_i, as does e; apply Theorem 1.8. ■

For $\{a_i : i \in I\} \subset G$, the intersection of all subgroups containing every a_i is the smallest such subgroup, the **subgroup generated by** $\{a_i\}$; if it is G, the a_i **generate** G.

Theorem 1.18 (7.6)

The subgroup H generated by $\{a_i : i \in I\}$ consists exactly of the finite products of integral powers of the a_i (a given a_i may occur several times).

Proof. Let K be the set of such finite products. K is closed under multiplication (juxtapose), contains $e = a_i^0$, and contains inverses: $(a_{i_1}^{n_1} \cdots a_{i_k}^{n_k})^{-1} = a_{i_k}^{-n_k} \cdots a_{i_1}^{-n_1}$. So $K \leq G$ contains each a_i, whence $H \subset K$. Conversely any subgroup containing the a_i contains all such products, so $K \subset H$. ■

A **Cayley digraph** of G for a generating set S has the elements of G as vertices and an arc of type s from g to gs for each $s \in S$. It is connected; at each vertex exactly one arc of each type starts and one ends; and two different paths from a vertex g ending at the same vertex describe equal products in G.

Example 1.4

$V = \{e, a, b, c\}$ is generated by $\{a, b\}$ but is not cyclic. $\mathbb{Z}_6$ is generated by $\{2, 3\}$, as $1 = 3 - 2$. The dihedral-like group with digraph two concentric triangles joined by spokes (generators a of order 3 and b of order 2 with $ba = a^2b$) is S_3.

1.7 Selected exercises

Motivation. Exercises: weaker group axioms, a finiteness criterion via subgroups, and orders of commuting elements (used again in §11).

Exercise 1.1 (§4, Ex. 39; groups via solvable equations)

Let G be a nonempty set with an associative binary operation $*$ such that for all $a, b \in G$ the equations $a * x = b$ and $y * a = b$ have solutions in G. Show that $\langle G, * \rangle$ is a group.

Solution.

Step 1. (A left identity.) Fix $c \in G$ and choose e with $e * c = c$. For $b \in G$ choose x with $c * x = b$. Then $e * b = e * (c * x) = (e * c) * x = c * x = b$. So $e * b = b$ for every b.

Step 2. (Left inverses.) For $a \in G$ choose a' with $a' * a = e$.

Step 3. (a' is also a right inverse.) Let a'' satisfy $a'' * a' = e$. Using Step 1 and associativity,

$$a * a' = e * (a * a') = (a'' * a') * (a * a') = a'' * ((a' * a) * a') = a'' * (e * a') = a'' * a' = e.$$

Step 4. (e is a two-sided identity.) $a * e = a * (a' * a) = (a * a') * a = e * a = a$.

So $\mathcal{G}_1$–$\mathcal{G}_3$ hold and G is a group. ■

Exercise 1.2 (§6, Ex. 48; finitely many subgroups)

Show that a group with only finitely many subgroups is finite.

Solution.

Step 1. Every $g \in G$ lies in the cyclic subgroup $\langle g \rangle$. There are only finitely many distinct cyclic subgroups $\langle g_1 \rangle, \dots, \langle g_k \rangle$, so $G = \langle g_1 \rangle \cup \cdots \cup \langle g_k \rangle$.

Step 2. Each $\langle g_i \rangle$ is finite. Otherwise $\langle g_i \rangle \cong \mathbb{Z}$ by Theorem 1.14, via $g_i^m \mapsto m$. The subgroups $n\mathbb{Z}$, $n = 1, 2, \dots$, of $\mathbb{Z}$ are distinct (the least positive element of $n\mathbb{Z}$ is n), so their preimages $\langle g_i^n \rangle$ are infinitely many distinct subgroups of G, a contradiction.

Step 3. Hence $|G| \le |\langle g_1 \rangle| + \cdots + |\langle g_k \rangle| < \infty$. ■

Exercise 1.3 (§6, Ex. 56; orders of commuting elements)

Let G be abelian and let $H, K \le G$ be cyclic of orders r and s.
(a) If $\gcd(r, s) = 1$, then G contains a cyclic subgroup of order rs.
(b) In general G contains a cyclic subgroup of order $\operatorname{lcm}(r, s)$.

Solution.

Step 1. (Key fact.) Let $a, b \in G$ have orders m, n with $\gcd(m, n) = 1$. We show ab has order mn. Since G is abelian, $(ab)^{mn} = (a^m)^n (b^n)^m = e$. Let t be the order of ab. Then $(ab)^t = e$ gives

$c := a^t = b^{-t} \in \langle a\rangle \cap \langle b\rangle$. By Theorem 1.15 the order of c divides m (as $c \in \langle a\rangle$) and divides n (as $c \in \langle b\rangle$), so it is 1 and $c = e$. Thus $a^t = e$ and $b^t = e$, so $m \mid t$ and $n \mid t$. Write $t = mt'$; then $n \mid mt'$ with $\gcd(n, m) = 1$ gives $n \mid t'$, so $mn \mid t$. Hence $t = mn$.

(a) Let $H = \langle a\rangle$, $K = \langle b\rangle$. By Step 1, $\langle ab\rangle$ is cyclic of order rs.

(b)

Step 2. Write $r = \prod_p p^{\alpha_p}$ and $s = \prod_p p^{\beta_p}$ over the primes p dividing rs, so $\operatorname{lcm}(r, s) = \prod_p p^{\max(\alpha_p,\beta_p)}$. Put

$$d_1 = \prod_{\alpha_p \geq \beta_p} p^{\alpha_p}, \qquad d_2 = \prod_{\alpha_p < \beta_p} p^{\beta_p}.$$

Then $d_1 \mid r$, $d_2 \mid s$, $\gcd(d_1, d_2) = 1$ and $d_1 d_2 = \operatorname{lcm}(r, s)$.

Step 3. By Theorem 1.15, a^{r/d_1} has order $r/\gcd(r, r/d_1) = d_1$, and likewise b^{s/d_2} has order d_2. By Step 1, $a^{r/d_1} b^{s/d_2}$ has order $d_1 d_2$ and generates a cyclic subgroup of order $\operatorname{lcm}(r, s)$. ■